

Machine Learning (CAI-262)

Objectives:

- K Nearest Neighbor Classifier
 - Selection of parameter k
 - Curse of dimensionality
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Classification:

The Classification algorithm is a Supervised Learning technique that is used to identify the category of new observations on the basis of training data. In Classification, a program learns from the given dataset or observations and then classifies new observation into a number of classes or groups. Such as, Yes or No, 0 or 1, Spam or Not Spam, cat or dog, etc. Classes can be called as targets/labels or categories.

The algorithm which implements the classification on a dataset is known as a classifier. There are two types of Classifications:

- **Binary Classifier:** If the classification problem has only two possible outcomes, then it is called as Binary Classifier.
Examples: YES or NO, MALE or FEMALE, SPAM or NOT SPAM, CAT or DOG, etc.
- **Multi-class Classifier:** If a classification problem has more than two outcomes, then it is called as Multi-class Classifier.
Example: Classifications of types of crops, Classification of types of music etc.

K Nearest Neighbor Classifier

K-Nearest Neighbor is one of the simplest Machine Learning algorithms based on Supervised Learning technique. It stores all the available data and classifies a new data point based on the similarity using following steps.

Step-1: Select the number K of the neighbors

Step-2: Calculate the Euclidean distance of K number of neighbors

Step-3: Take the K nearest neighbors as per the calculated Euclidean distance.

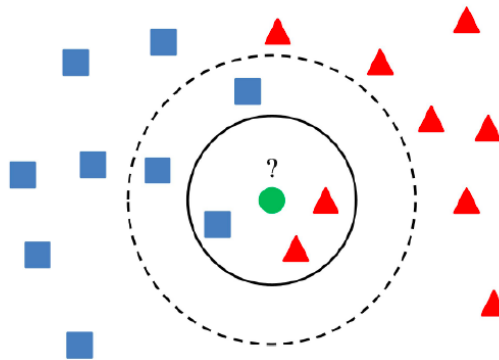
Step-4: Among these k neighbors, count the number of the data points in each category.

Step-5: Assign the new data points to that category for which the number of the neighbor is maximum.

The main idea behind K-NN is to find the K nearest data points to a given data point and then predict the label of the given data point based on the labels of its K nearest neighbors.

Find distance of test point with all training points

What would be class of green point if we consider $K=3$? And $K=5$? So the parameter K is very important for training the data.



Example:

To solve the numerical example on the K-nearest neighbor i.e. KNN classification algorithm, we will use the following dataset.

Point	Coordinates	Class Label
A1	(2,10)	C2
A2	(2, 6)	C1
A3	(11,11)	C3
A4	(6, 9)	C2
A5	(6, 5)	C1
A6	(1, 2)	C1
A7	(5, 10)	C2
A8	(4, 9)	C2

A9	(10, 12)	C3
A10	(7, 5)	C1
A11	(9, 11)	C3
A12	(4, 6)	C1
A13	(3, 10)	C2
A14	(3, 8)	C2
A15	(6, 11)	C2

In the above dataset, we have fifteen data points with three class labels. Now, suppose that we have to find the class label of the point P= (5, 7).

Point	Coordinates	Distance from P (5, 7)
A1	(2, 10)	4.24
A2	(2, 6)	3.16
A3	(11, 11)	7.21
A4	(6, 9)	2.23
A5	(6, 5)	2.23
A6	(1, 2)	6.40
A7	(5, 10)	3.0
A8	(4, 9)	2.23
A9	(10, 12)	7.07
A10	(7, 5)	2.82
A11	(9, 11)	5.65
A12	(4, 6)	1.41
A13	(3, 10)	3.60
A15	(3, 8)	2.23

A15	(6, 11)	4.12
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Sorted Points according to distances are given below.

Point	Coordinates	Distance from P (5, 7)
A12	(4, 6)	1.41
A4	(6, 9)	2.23
A5	(6, 5)	2.23
A8	(4, 9)	2.23
A15	(3, 8)	2.23
A10	(7, 5)	2.82
A7	(5, 10)	3
A2	(2, 6)	3.16
A13	(3, 10)	3.6
A15	(6, 11)	4.12
A1	(2, 10)	4.24
A11	(9, 11)	5.65
A6	(1, 2)	6.4
A9	(10, 12)	7.07
A3	(11, 11)	7.21

For this, we will first specify the number of nearest neighbors i.e. k . Let us take k to be 3. Now, we will find the distance of P to each data point in the dataset. For this KNN classification numerical example, we will use the euclidean distance metric. The following table shows the euclidean distance of P to each data point in the dataset.

Distance of each point for the new point

After finding the distance of each point in the dataset to P, we will sort the above points according to their distance from P (5, 7). After sorting, we get the following table.

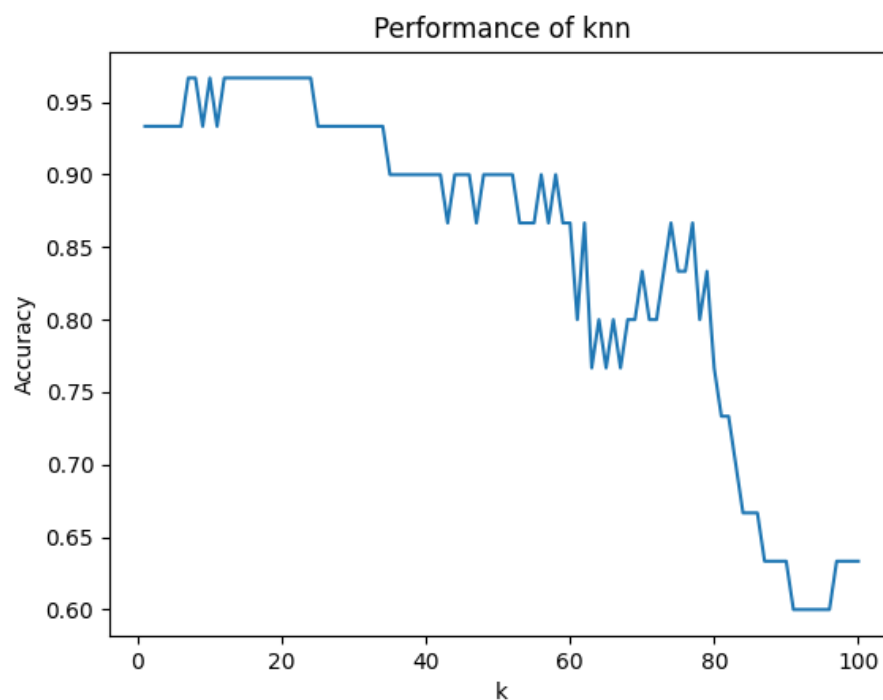
As we have taken $k=3$, we will now consider the class labels of three points in the dataset nearest to point P to classify P. In the above table, A12, A4, and A5 are the closest 3 neighbors of point P. Hence, we will use the class labels of points A12, A4, and A5 to decide the class label for P.

Now, point A12, A4, and A5 have the class labels C1, C2, and C1 respectively. Among these points, the majority class label is C1. Therefore, we will specify the class label of point P = (5, 7) as C1. Hence, we have successfully used KNN classification to classify point P according to the given dataset.

By studying the above KNN classification numerical example, you can see that the algorithm is straight forward and doesn't require any specific mathematical skills apart from distance calculation and majority selection.

Selection of parameter K

We can find an optimal value of k by iterating through a range of k and plotting the performance of the model.



Advantages of KNN Algorithm:

- It is simple to implement.
- It is robust to the noisy training data.
- It can be more effective if the training data is large.

Disadvantages of KNN Algorithm:

- Always needs to determine the value of K which may be complex some time.
- The computation cost is high because of calculating the distance between the data points for all the training samples.

Curse of Dimensionality:

The curse of dimensionality refers to various phenomena that arise when analyzing and organizing data in high-dimensional spaces that do not occur in low-dimensional settings. The expression was coined by Richard E. Bellman. The curse generally refers to issues that arise when the number of datapoints is small relative to the dimension of the data.